

Apply filters to SQL queries

Project description

In this project, I demonstrate how SQL filters can be used to investigate login attempts and employee data. By applying logical and comparison operators, I identify failed logins, specific dates, and department-based employee records. These queries showcase how filtering helps cybersecurity analysts isolate relevant information efficiently.

Retrieve after hours failed login attempts

```
SELECT *  
FROM log_in_attempts  
WHERE success = 0 AND login_time > '18:00:00';
```

Explanation: This query returns all failed login attempts (`success = 0`) that occurred after 6 PM. It helps identify potential unauthorized access attempts outside normal working hours.

Retrieve login attempts on specific dates

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-08' OR login_date = '2022-05-09';
```

Explanation: Filters login attempts that happened on May 8 or May 9, 2022. The **OR** operator ensures both dates are included.

Retrieve login attempts outside of Mexico

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE '%MEX%';
```

Explanation: Excludes all login attempts from Mexico (including “MEX” or “MEXICO”). The **NOT** operator reverses the condition to show only other countries.

Retrieve employees in Marketing

```
SELECT *  
FROM employees  
WHERE department LIKE '%Marketing%' AND office LIKE 'East%';
```

Explanation: Finds employees who work in the Marketing department and are located in the East building. The **AND** operator ensures both conditions are met.

Retrieve employees in Finance or Sales

```
SELECT *  
FROM employees  
WHERE department LIKE '%Finance%' OR department LIKE '%Sales%';
```

Explanation: Returns employees who belong to either the Finance or Sales departments. The **OR** operator includes both groups.

Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE department NOT LIKE '%Information Technology%';
```

Explanation: Filters out employees from the IT department. The **NOT** operator excludes matching rows.

Summary

These queries demonstrate how SQL filtering operators — **AND**, **OR**, **NOT**, and comparison operators — can be combined to extract precise subsets of data. They are essential for cybersecurity analysis, allowing professionals to quickly identify anomalies, trends, and relevant records.